

Setup & Maintenance Manual

AssetCap Server - EduCloud OpenStack / Ubuntu 24.04

Updated after migration cutover to the new OpenStack server

Item	Current value
Primary hostname	bops-assetcap-er8p
Platform	UBC EduCloud (OpenStack / Horizon)
Operating system	Ubuntu 24.04 LTS (Noble)
Primary Linux admin user	developer
Web stack	Nginx + Gunicorn + systemd
Patch management	AWS Systems Manager (SSM) agent via snap
Document status	Updated for post-migration operations

Important: this manual preserves the known application layout from the previous server, updates the OS / OpenStack / patching model for the new server, and flags any values that should be re-verified with UBC IT if the SSH endpoint or project assignment changes.

1. Environment overview

The prior manual documented the AssetCap application stack, service names, domains, ports, and file paths on the older Ubuntu 22.04 server. **The migrated environment now runs on Ubuntu 24.04 in UBC EduCloud OpenStack.** The application topology remains the same, while infrastructure administration now aligns with the OpenStack / Horizon operating model and SSM-based OS patching.

Component	Current state / note
Virtualization platform	EduCloud OpenStack (on-prem private cloud)
OpenStack portal	https://educloud.ubc.ca
Portal access	Restricted to UBC campus IP space, MFA enabled, access managed through Entra ID group
OpenStack portal type	Horizon web management portal
API access	Not available at this time per OpenStack 101 training
Migration cleanup	Temporary migration account migration_admin removed after cutover

Recommended access checks

- Confirm your account is in the correct Entra ID group for the project.
- Use MFA and connect from campus IP space or an approved access path.
- Record the current SSH endpoint separately if IT provides a specific public IP or DNS name for remote administration.
- Use the OpenStack portal for VM management tasks; application administration remains on the Ubuntu host.

2. Administrative access and key commands

Task	Command / method	Notes
SSH login	<code>ssh developer@<current-server-ip-or-dns></code>	Replace with the current endpoint supplied by IT if it differs from the old server.
OpenStack portal	<code>https://educloud.ubc.ca</code>	Use browser access with MFA; membership controlled by Entra ID group.
Check OS version	<code>lsb_release -a</code>	Should report Ubuntu 24.04 LTS (Noble).
Check hostname	<code>hostnamectl</code>	Expected host: bops-assetcap-er8p.
Disk overview	<code>df -h</code>	Useful for general storage checks before uploads or OCR processing.
Service logs	<code>sudo journalctl -u <service-name> -n 100 --no-pager</code>	Use -f for live follow mode.

PostgreSQL access examples

```
# SSH tunnel from local machine (replace endpoint if needed)
ssh -L 5433:127.0.0.1:5432 developer@<current-server-ip-or-dns>

# Backup from the Ubuntu host
pg_dump -U developer -h localhost qr_code_db > backup_qr_code.sql
```

Authentication service utility commands

```
cd /home/developer/auth_service
python3 -m venv venv
source venv/bin/activate
python3 init_db.py <username> <email>
python reset_password.py <username> <new_password>
```

3. Application inventory

The following application stack is carried forward from the previous manual and remains the expected runtime layout unless you have intentionally changed a service definition, port, or Gunicorn target.

Port	Domain	systemd service	Path	Gunicorn target
8000	appprod.assetcap.facilities.ubc.ca	assetcap-app.service	/home/developer/asset_capture_app_dev	asset_plate_reviewer:app
8001	reviewme.assetcap.facilities.ubc.ca	assetcap-reviewme	/home/developer/review/Asset_dashboard_browser_ME	asset_plate_reviewer:app
8002	dashboardprod.assetcap.facilities.ubc.ca	assetcap-dashboard	/home/developer/Dashboard	Asset_portal_dashboard:app
8003	sdiprocess.assetcap.facilities.ubc.ca	sdi_process	/home/developer/SDI_processor	app:app

Port	Domain	systemd service	Path	Gunicorn target
	ties.ubc.ca		ss	
8004	reviewbf.assetcap.facilities.ubc.ca	assetcap-bf	/home/developer/review/Asset_dashboard_browser_BF	asset_plate_reviewer_bf:app
8005	reviewel.assetcap.facilities.ubc.ca	assetcap-el	/home/developer/review/Asset_dashboard_browser_EL	Asset_dashboard_EL:app

Restart commands

```
sudo systemctl restart assetcap-app.service
sudo systemctl restart assetcap-reviewme
sudo systemctl restart assetcap-dashboard
sudo systemctl restart sdi_process
sudo systemctl restart assetcap-bf
sudo systemctl restart assetcap-el
```

4. Key directories and data paths

```
/home/developer/
|- SDI_process/
|  |- venv/
|  |- app.py
|  |- static/
|  `-- template/
|- Dashboard/
|  |- venv/
|  |- Asset_portal_dashboard.py
|  `-- charts/approval.py
|- review/Asset_dashboard_browser_ME/
|  `-- asset_plate_reviewer.py
|- review/Asset_dashboard_browser_BF/
|  `-- asset_plate_reviewer_bf.py
|- review/Asset_dashboard_browser_EL/
|  `-- Asset_dashboard_EL.py
|- Output_jason_api/
`-- Capture_photos_upload/
```

- SQLite checkpoint helper: /home/developer/asset_capture_app_dev/data/sqlite_checkpoint.sh
- Pip cache cleanup: rm -rf /home/developer/.cache/pip
- Check listening ports with sudo ss -ltnp | grep "<port>" before troubleshooting Nginx or Gunicorn issues.

5. Logs, debugging, and health checks

Check	Command
Live SDI logs	sudo journalctl -u sdi_process -f
Dashboard last 100 lines	sudo journalctl -u assetcap-dashboard -n 100 --no-pager

Check	Command
Nginx global error log	sudo tail -n 100 /var/log/nginx/error.log
Dashboard Nginx site log	sudo tail -n 100 /var/log/nginx/dashboardprod.error.log
Port 8003 listening	sudo ss -ltnp grep ":8003"
Service health summary	systemctl --type=service --state=running grep assetcap

Useful host checks

```
hostnamectl
lsb_release -a
df -h
free -h
sudo systemctl status snap.amazon-ssm-agent.amazon-ssm-agent --no-pager
snap services amazon-ssm-agent
```

6. OS patching and compliance model

This server is enrolled in the new **AWS Systems Manager (SSM)** patching model used for Hybrid Cloud workloads. The current Ubuntu host uses the snap-based SSM agent service, not a traditional `amazon-ssm-agent.service` unit.

Item	Current guidance
Agent package	amazon-ssm-agent installed via snap
Correct status check	<code>sudo systemctl status snap.amazon-ssm-agent.amazon-ssm-agent --no-pager</code>
Alternative status check	<code>snap services amazon-ssm-agent</code>
Primary log file	<code>sudo tail -n 200 /var/log/amazon/ssm/amazon-ssm-agent.log</code>
Observed region	ca-central-1
Observed compliance result	Instance reported compliant during patch policy execution
Reboot behavior	Patch baseline is configured as <code>RebootIfNeeded</code> when required

Important operational notes

- Do not assume you should disable unattended-upgrades on your own. That decision may be governed by Cybersecurity / Hybrid Cloud standards.
- If SSM logs show `dpkg` or `apt` lock contention, raise it with the Hybrid Cloud Service team before changing patch automation settings locally.
- A prior patch run showed successful compliance but also logged `S3 AccessDenied` on patch output upload; this is an IAM / policy issue for the cloud team, not a local Ubuntu failure.
- Check for reboot-required status with: `test -f /var/run/reboot-required && cat /var/run/reboot-required || echo "No reboot required"`.

7. Deployment flow after code updates

- Go to the application folder you are updating.
- Activate the app-specific virtual environment.
- Install or refresh requirements if needed.
- Restart only the affected `systemd` service.
- Validate service status, Gunicorn logs, and Nginx routing before closing the change.

```
cd /home/developer/SDI_process
source venv/bin/activate
pip install -r requirements.txt
deactivate
sudo systemctl restart sdi_process
sudo journalctl -u sdi_process -n 50 --no-pager
```

8. Base system and Python package references

System packages typically present on the host

```
sudo apt update
sudo apt install -y python3 python3-venv python3-pip \
    nginx certbot python3-certbot-nginx \
    tesseract-ocr libtesseract-dev \
    libgl1
```

Python packages commonly used in app virtual environments

- Common: flask, gunicorn, python-dotenv
- SDI process: pandas, openpyxl
- Reviewer / OCR apps: openai, opencv-python, pillow, pytesseract, numpy, httpx, pydantic and supporting dependencies

9. OpenStack administrator notes

- OpenStack terminology differs from VMware: Cloud Director -> Horizon, Organization -> Project, VMs -> Instances, Snapshot -> Instance Snapshot.
- The portal is intended for infrastructure-level tasks such as viewing instances, projects, limits, and snapshots.
- Application owners still manage Flask code, Gunicorn services, Nginx, databases, and local file structure from the Ubuntu host.
- If project membership or dashboard access fails, request confirmation of the assigned Entra ID group and any campus-IP access requirements.

10. Post-migration cleanup and verification

Verification item	Command / check
Migration user removed	id migration_admin # should return no such user
SSM agent active	snap services amazon-ssm-agent
No broken apt state	sudo apt -f install
No reboot pending	test -f /var/run/reboot-required && cat /var/run/reboot-required echo "No reboot required"
Domain routing	Validate all production domains through browser / curl after restart
Portal access	Open https://educcloud.ubc.ca and confirm MFA + project visibility

Document maintenance note: if IT later confirms a new SSH endpoint, project name, or patching standard, update this manual immediately so the operational runbook stays aligned with the OpenStack environment.